

Alaaeldin El-Nouby

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Education

École Normale Supérieure and Meta AI

Paris, France

PhD in Computer Science

August 2020 - August 2023

- Advisors: Hervé Jegou, Ivan Laptev, and Natalia Neverova.
- Committee: Alyosha Efros, Cees Snoek, Paolo Favaro, Tinne Tuytelaars
- Research: Vision Transformers, Self-Supervised Learning, Language-Vision Pre-training, Image Compression, Image Retrieval.

University of Guelph

Guelph, Ontario, Canada

MASc in Engineering Systems and Computing

September 2017 - September 2019

- Advisors: Graham Taylor, Rozita Dara
- GPA: 4.0
- Awards: Vector Institute Research Grant, CEPS Dean's Scholarship

Ain Shams University

Cairo, Egypt

BEng Computers and Systems Engineering

September 2011 - June 2016

- Advisor: Hazem Abbas
- Grade: Distinction

Experience

Meta Superintelligence Labs

Paris, France

Research Scientist (IC6)

June 2025 - Present

- Co-led LLM pre-training for Code World Models (CWMv2) including architecture design, optimization, and data.
- Working as part of the post-training team driving the agentic coding capabilities of Muse Spark, Meta's frontier model.

MLR, Apple Inc.

Paris, France

Senior Research Scientist

August 2023 - June 2025

- Led a team that conducted research in large-scale pre-training of vision-only and multimodal models. Notably, we released multiple research papers and model checkpoints for our work on Autoregressive Image Models (i.e., AIMv1, AIMv2). Moreover, we published large-scale scaling laws studies for native multimodal models (ICCV25 Oral) and scaling laws for data mixing.

FAIR, Meta AI

Paris, France

PhD Resident

August 2020 - August 2023

- Advised by Hervé Jegou and Natalia Neverova at FAIR. Conducted research on Vision Transformers, Self-supervised learning, Language-vision pre-training, Image Retrieval and Image Compression. My research at FAIR resulted in 12 publications, with 6 as the primary author.

Apple Inc.

Cupertino, CA, US

Research Intern

February 2019 - May 2019

- Worked as a part of the Video Engineering Group under the supervision of Josh Susskind and Shuangfei Zhai, conducting research on video understanding. An early version of this project won the "Best Poster Award" at the *Large Scale Holistic Video Understanding workshop, ICCV2019*.

Microsoft Research

Montreal, Canada

Research Intern

August 2018 - November 2018

- Worked on using generative models and dialogues for text-to-image synthesis in a sequential multi-step manner. The internship resulted in GeNeVA, a novel image generation method published at *ICCV 2019*.

Microsoft ATL| NLP Team

Cairo, Egypt

Research Software Development Engineer

August 2016 - August 2017

- Worked on developing a task oriented virtual agent for customer support. My focus was on developing engineering pipelines that automate the extraction of answers from different sources on the web (e.g., web pages, click through logs).

Academic Service

Reviewing	CVPR'22-'25, ECCV'22-'24, NeurIPS'24, ICLR'24, ICCV'21, TPAMI (2021-Present).
Teaching	CIS*3110 Operating Systems - <i>Uni. of Guelph</i> , CSE*121 Fundamentals of Programming - <i>Ain Shams University</i>
Organization	The Workshop on Transformers for Vision (T4V) - 2024 & 2025
Interns	Mustafa Shukor (Apple), Federico Baldassarre (Meta, Co-advised with Hervé Jegou), Joao Maria Janeiro (Meta, Co-advised with Jakob Verbeek)

Invited Talks

- **L3D-IVU Workshop (invited talk) - CVPR'24** Scalable Pre-training of Large Autoregressive Image Models
- **IMAGINE lab - École des Ponts ParisTech** Bringing the power of Transformers to Computer Vision
- **Vector Institute / University of Guelph** - Masked Image Modeling for Visual Representation Learning
- **Transformers for Vision workshop, CVPR'22** - Are Large-scale Datasets Necessary for Self-Supervised Pre-training?
- **Max Planck Institute / Tübingen AI Center** - Are Large-scale Datasets Necessary for Self-Supervised Pre-training?
- **Computer Vision Group (CVG), University of Bern** - Are Large-scale Datasets Necessary for Self-Supervised Pre-training?
- **Large Scale Holistic Video Understanding workshop, CVPR'21** - Training Vision Transformers for Image Retrieval
- **KTH Royal Institute of Technology** - Training Vision Transformers for Image Retrieval
- **Microsoft Research Montreal** - Sequential Scene Understanding and Generation
- **DeepVision workshop, Simon Fraser University** - Real-Time End-to-End Action Detection with Two-Stream Networks
- **Twenty Billion Neurons** - Real-Time End-to-End Action Detection with Two-Stream Networks

In The Press

- Discussed IMAGEBIND during an appearance on a popular talk show on Egyptian television [Interview]
- IMAGEBIND [Announcement by Mark Zuckerberg, The Verge, Engadget, WorldOfAI]
- DINOv2 [Announcement by Mark Zuckerberg, Venture Beat, KD Nuggets, AI Papers Academy]
- Yannic Kilcher covered and explained XCiT on his channel. [Video].
- Multiple blog posts have been published discussing and explaining XCiT [AI Summer, Veritable Tech Blog, Erik Storrs Blog]

Selected Publications

- Scaling Laws for Native Multimodal Models - Mustafa Shukor, Enrico Fini, Victor Guilherme Turrissi da Costa, Matthieu Cord, Joshua M Susskind, [Alaaeldin El-Nouby](#). ICCV 2025 - Oral
- FlexTok: Resampling Images into 1D Token Sequences of Flexible Length - Roman Bachmann, Jesse Allardice, David Mizrahi, Enrico Fini, Oğuzhan Fatih Kar, Elmira Amirloo, [Alaaeldin El-Nouby](#), Amir Zamir, Afshin Dehghan. ICML 2025
- Parameters vs flops: Scaling laws for optimal sparsity for mixture-of-experts language models - Samira Abnar, Harshay Shah, Dan Busbridge, [Alaaeldin El-Nouby](#), Josh Susskind, Vimal Thilak. ICML 2025
- Multimodal Autoregressive Pre-training of Large Vision Encoders - Enrico Fini*, Mustafa Shukor*, Xiujuan Li, Philipp Dufter, Michal Klein, David Haldimann, Sai Aitharaju, Victor Guilherme Turrissi da Costa, Louis Béthune, Zhe Gan, Alexander T Toshev, Marcin Eichner, Moin Nabi, Yinfei Yang, Joshua M Susskind, [Alaaeldin El-Nouby*](#) CVPR 2025 - Highlight
- Scalable Pre-training of Large Autoregressive Image Models - Alaaeldin El-Nouby, Michal Klein, Shuangfei Zhai, Miguel Angel Bautista, Alexander Toshev, Vaishaal Shankar, Joshua M Susskind, Armand Joulin. ICML 2024
- IMAGEBIND: One Embedding Space To Bind Them All - Rohit Girdhar*, [Alaaeldin El-Nouby*](#), Zhuang Liu, Mannat Singh, Kalyan Vasudev Alwala, Armand Joulin, Ishan Misra*. CVPR 2023
- OmniMAE: Single Model Masked Pre-training on Images and Videos - Rohit Girdhar*, [Alaaeldin El-Nouby*](#), Mannat Singh*, Kalyan Vasudev Alwala*, Armand Joulin, Ishan Misra*. CVPR 2023 - Highlight
- Image Compression with Product Quantized Masked Image Modeling - [Alaaeldin El-Nouby](#), Matthew J. Muckley, Karen Ullrich, Ivan Laptev, Jakob Verbeek, Hervé Jégou. TMLR 2023
- Are Large-scale Datasets Necessary for Self-Supervised Pre-training? - [Alaaeldin El-Nouby*](#), Gautier Izacard*, Hugo Touvron, Ivan Laptev, Hervé Jégou, Edouard Grave.
- XCiT: Cross-Covariance Image Transformers - Alaaeldin El-Nouby, Hugo Touvron, Mathilde Caron, Piotr Bojanowski, Matthijs Douze, Armand Joulin, Ivan Laptev, Natalia Neverova, Gabriel Synnaeve, Jakob Verbeek, Hervé Jégou. NeurIPS 2021
- LeViT: a Vision Transformer in ConvNet's Clothing for Faster Inference - Ben Graham, [Alaaeldin El-Nouby](#), Hugo Touvron, Pierre Stock, Armand Joulin, Hervé Jégou, Matthijs Douze. ICCV 2021
- Tell, Draw, and Repeat: Generating and Modifying Images Based on Continual Linguistic Instruction - [Alaaeldin El-Nouby](#), Shikhar Sharma, Hannes Schulz, Devon Hjelm, Layla El Asri, Samira Ebrahimi Kahou, Yoshua Bengio, Graham W Taylor. ICCV 2019